

GLOBAL INDUSTRY CHALLENGE 2026

TEAM COVER PAGE

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Device-Native Dirac-3 Optimization for Cost and Resilience in Hierarchical Microgrids

Team Quantum Pattern Recognition — Global Industry Challenge 2026, QCi track: Cost Optimization in Resilient Power Grids

Abstract

We present a reproducible three-stage hybrid pipeline for investment, contingency islanding, and nonlinear dispatch on public IEEE-39 transmission and IEEE-33 radial-feeder instances. The IEEE-39 planning layer assigns every bus to one of 14 connected, pairwise-disjoint zones and preserves all scenario probability. One here-and-now stochastic master fixes portfolio $\{0,3\}$ before evaluating 15 retained scenarios. Expected total annual cost, including USD 221.23M/year of upgrades, falls from USD 2.486B to USD 2.242B (9.80%); maximum load-weighted system fraction unserved per hour falls from 0.1932 to 0.0672, and expected critical outage hours from 0.080 to 0.000. The selected zones contain 4/21 load buses and 27.31% of system MW. The IEEE-33 exact frontier contains three policies: USD 13.42M/21.0, USD 16.44M/4.2, and USD 19.46M/0.0 expected annual critical energy not served (MWh/year). One credential-free command regenerates both studies, 15 IEEE-39 convention/invariant gates, 21 unit tests, and 39/39 consolidated claims. All 27 feeder payloads and five registered IEEE-39 dispatch payloads pass coefficient-spread and distinct-separation checks. The frozen Dirac-3 campaign returned 250/250 requested samples across 10/10 audited jobs; all 9/9 in-spec jobs passed immutable configuration and raw-feasibility checks, producing 225 evidence samples in 46 device seconds. Exact optima were recovered for IEEE-33 Stages 1, 2, and native cubic-qudit Stage 3; the matched continuous gap was 0.00378%. IEEE-39 best gaps were 0.00606%-1.728% hourly and 1.039% for the certified 13-variable result. The separately labelled 121-variable probe returned samples but remains outside coefficient-resolution guidance and is not optimization evidence.

1. Problem, evidence boundary, and contributions

The challenge asks for concrete execution of a resilient-grid optimization pipeline: decide what to build, choose operable islands under faults, and dispatch resources while reporting cost, customer service, and critical-load continuity [1]. We use the public IEEE-39 New England case [4] as a transmission-level planning and dispatch stress test and aggregate the Baran-Wu IEEE-33 feeder [3] into four connected service sections. Synthetic DER prices, event rates, critical labels, load/PV shapes, loss factors, and value-of-lost-load assumptions are disclosed in source; they are demonstration assumptions, not utility data or operational advice. Related quantum-energy work addresses microgrid energy management, resilient storage topology, and unit commitment [5-7]; our focus is verifiable device-native polynomial execution within a fail-closed evidence chain.

Our contribution is device-envelope engineering. The model uses Dirac-3 integer/qudit and continuous-simplex encodings directly, including repeated indices for cubic terms. Every registered payload is audited against degree-dependent variable limits, the 949-total-level limit, coefficient spread, and pairwise distinct-coefficient separation [2]. Constants omitted by the device are restored during local float64 scoring. A raw sample must satisfy its registered integer levels or non-negative simplex and balance tolerance; this machine-domain pass is distinct from equipment-cap and chronological feasibility. A separate physical-decode audit reports those limits, and projection, clamping, or redispatch may diagnose implications but cannot create a hardware pass. Device results are compared with exact enumeration, HiGHS MILP, deterministic one-dimensional optimization, an analytic convex-simplex KKT oracle, and seeded simulated annealing.

2. Hierarchical formulation

Hybrid quantum-classical siting, islanding and dispatch — system planning is exact/HiGHS; device-native dispatch objectives run on QCi Dirac-3

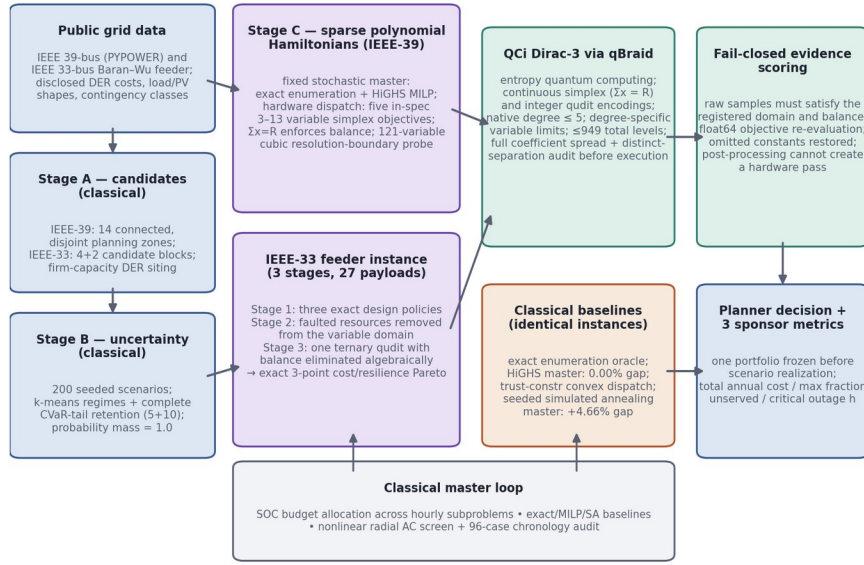


Figure 1. Fixed stochastic planning is classical; QCi Dirac-3 executes or benchmarks the registered dispatch and feeder polynomial subproblems.

For IEEE-39, a deterministic multi-source electrical-distance partition assigns every bus to exactly one of 14 connected zones. Candidate loads sum to the 6,254.23 MW system load and pairwise overlap is zero. Two hundred seeded scenarios use the complete PCC identifier domain; compression retains five regime representatives and all ten CVaR-tail cases, so probability remains 1.0. Binary upgrade variables are chosen once by a classical two-stage stochastic master that minimizes expected 24 h energy-plus-VoLL cost plus one day of annualized upgrade cost. The same portfolio is then evaluated in every scenario without redesign. The QCi IEEE-39 runs address disclosed dispatch benchmark objectives, not this planning master. IEEE-33 uses four primary sections plus two physically distinct black-start overlaps. A Stage-0 exact screen reports the sectionalization/capacity-margin frontier before any Hamiltonian is constructed.

For feeder design, binary item variables x encode a feasible product at a candidate section. The Stage-1 Hamiltonian is annualized upgrade cost plus beta times frequency-weighted critical energy not served, with exactly-one constraints for primary sections and at-most-one constraints for overlaps. Cost and risk therefore share USD/year units. Exact optimization at $\beta = 0, 30,000$, and $75,000$ USD/MWh produces Plans A, B, and C. In Stage 2, a faulted candidate is not a legal decision: it is structurally removed. Available island variables a minimize $H_2 = -\sum_i (V_i - 100) a_i$ plus overlap exclusion terms, where V_i is four-hour firm/critical service value. This avoids large analog penalties for physical impossibilities.

Stage 3 uses cubic wear/conversion curves $f_k(p) = c_{1,k} p + c_{2,k} p^2 + c_{3,k} p^3$. For a two-resource pocket of load L , the native ternary qudit z in $\{0,1,2\}$ sets $p_{\text{primary}} = (z+1)L/3$ and $p_{\text{backup}} = (2-z)L/3$. Thus $p_{\text{primary}} + p_{\text{backup}} = L$ for every hardware state, with no balance penalty, and substitution produces a genuine rank-3 one-variable Hamiltonian. One ternary qudit represents the three valid states directly; two binary bits introduce an unused fourth state, while one-hot encoding requires three bits plus a constraint. A matched continuous payload uses shares $u_0 + u_1 = 1$ through the machine sum constraint and the same cubic curves; its deterministic golden-section oracle places the primary share within one percentage point of $2/3$. The exact qudit optimum is $z=1$.

The IEEE-39 stochastic master is checked by exact enumeration and HiGHS, which return the same fixed portfolio. Registered hardware objectives are four stressed hourly dispatches (3-4 variables) and one four-hour window (13 variables), each using the simplex equality as balance. The 13-variable result is a certified matched-objective result for a four-hour total-energy relaxation, not a certificate of hourly physical dispatch.

3. Credential-free results and physical audits

Instance / policy	Upfront cost	Expected annual critical ENS	Static critical pocket-hours	Worst static customer-block unserved
IEEE-33 Plan A — cost-efficient	USD 13.42M	21.0 MWh/year	16	34.375%
IEEE-33 Plan B — balanced-critical	USD 16.44M	4.2 MWh/year	8	25.0%
IEEE-33 Plan C — robust-critical	USD 19.46M	0.0 MWh/year	0	0.0%

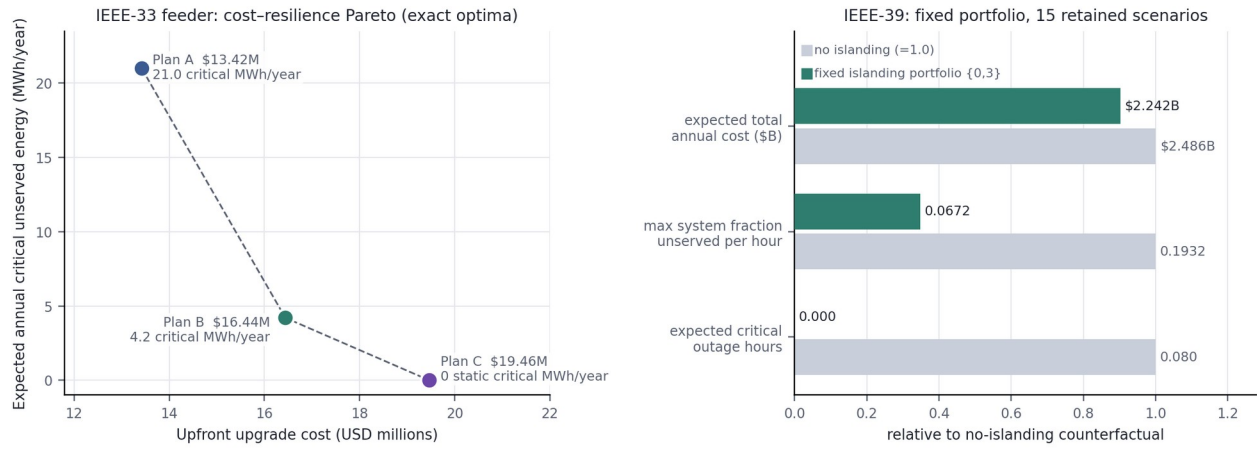


Figure 2. Left: exact IEEE-33 three-point cost-resilience frontier. Right: IEEE-39 expected total annual cost, maximum unserved fraction, and expected critical outage hours for the fixed portfolio.

IEEE-33 Pareto frontier and IEEE-39 sponsor metrics. The middle policy is non-dominated: a USD 3.02M increment over Plan A protects the higher-risk critical pocket and reduces expected annual critical energy not served by 80%. Plan C costs USD 6.04M more than Plan A and eliminates static critical shortfall in all four prototype contingencies. On IEEE-39, every load bus has exactly one planning-zone owner and the fixed master selects {0,3} before scenario realization. Expected 24 h energy-plus-VoLL cost falls from USD 6.811M to USD 5.537M; after annualization and addition of the USD 221.23M/year upgrade, expected total annual cost falls from USD 2.486B to USD 2.242B, saving USD 243.70M/year (9.80%). Maximum load-weighted system fraction unserved per hour falls from 0.1932 to 0.0672 and expected critical outage hours from 0.080 to 0.000. The selected zones contain 4/21 load buses and 27.31% of system MW. HiGHS matches exact enumeration; seeded simulated annealing has a 4.66% master-objective gap.

Static capacity is not energy readiness. A dependency-free nonlinear backward/forward-sweep AC screen converges for the grid-connected and every served-island case; base-case minimum voltage is 0.9131 p.u. and all screened islands remain within 0.90-1.10 p.u. A separate 96-case, 24-hour SOC/black-start audit varies four contingencies and six start times. It finds no power, SOC, reserve, or recursion violation, but conservative initial SOC exposes 0.877 MWh worst-case critical shortfall for static Plan C. We report this limitation rather than relabel Plan C as chronologically robust. A separately labelled 2 MW/8 MWh hardening sensitivity reduces worst critical shortfall to zero but still sheds 1.488 MWh of noncritical energy; it is not included in the three-point static frontier.

4. Frozen Dirac-3 experiment and scoring

The live experiment is frozen before device access. Schedule 1 is used for ten scientifically distinct jobs at 25 returned samples each: four IEEE-39 hourly objectives, one IEEE-39 window, the characterization probe, IEEE-33 Stage 1 balanced design, compound-fault Stage 2, and matched Stage-3 cubic-qudit and continuous objectives. One separate three-sample cubic-qudit smoke job checks credentials and allocation. Nine jobs are registered evidence; the probe remains characterization regardless of outcome. The system portfolio comes from the classical stochastic master; Dirac-3 solves or benchmarks the registered dispatch and feeder subproblems. Dirac-returned samples are not gate-QPU shots, and qBraid Lab compute allowance is distinct from QCi Dirac allocation or metering.

Live evidence group	Jobs	Samples/ job	Evidence class	Measured result
IEEE-39 hourly dispatch h16-h19	4	25	registered, in-spec	4/4 valid; n=100; 19 s; best gap 0.006064%-1.7281%
IEEE-39 four-hour window	1	25	registered, in-spec	1/1 valid; n=25; 9 s; best gap 1.0387%
IEEE-39 cubic coefficient probe	1	25	characterization only	1/1 valid; n=25; 274 s; out-of-guidance
IEEE-33 Stages 1-2	2	25	registered, in-spec	2/2 valid; n=50; 10 s; best gap 0%
IEEE-33 matched Stage 3	2	25	registered, in-spec	2/2 valid; n=50; 8 s; best gap 0%-0.0037756%

Submission requires an exact confirmation phrase. Job IDs are written immediately to immutable receipts; completed, errored, cancelled, and timed-out attempts are retained, never overwritten or silently retried. The frozen-ledger discipline follows [8]. The audit requires the payload hash, remote job type, schedule, requested and counted samples, job ID, device usage seconds, status, and pre/post allocation. It scores each raw feasible vector against the identical polynomial using float64 arithmetic and restores local constants. Smoke files cannot enter the evidence ledger. All 250 requested campaign samples were returned and every raw state passed its registered domain and balance check. The nine in-spec jobs used 46 device seconds. IEEE-33 Stages 1, 2, and cubic-qudit Stage 3 reached their exact ground states; the matched continuous Stage-3 gap was 0.00378%. The four IEEE-39 hourly best gaps ranged from 0.00606% to 1.728%. For the 13-variable four-hour total-energy relaxation, an analytic KKT/water-filling oracle certifies a 1.039% best gap over 25 feasible samples. The 121-variable cubic probe returned 25 samples in 274 s, but its $5.22e8$ coefficient spread fails the documented resolution guidance; it is hardware-boundary characterization, not solution-quality evidence. Immutable receipt timestamps give 138.330 s summed submit-to-collect wall time for the nine evidence jobs and 427.613 s from first submission to final collection for the sequential ten-job campaign; QCi reports 46 and 320 device seconds, respectively. The ledger, raw responses, job IDs, and deterministic certification are packaged under results/live.

The strict post-run audit independently re-read the immutable responses, receipts, and returned remote configurations: all 250/250 campaign samples pass their registered machine domains. That result is deliberately separated from physical dispatch. Across the four IEEE-39 hourly objectives, 72/100 raw samples are within calibrated source caps, and only 1/4 machine-objective best states is cap-compliant. Best-state overruns are 17.903 MW (h17 storage discharge), 2.127 MW (h18 legacy generation), and 2.102 MW (h19 microturbine); h16 has 0 MW. The 13-variable window enforces only total four-hour energy adequacy (aggregate residual 0.000148 MWh); its best state has a 45.425 MW maximum cap overrun and a 300.305 MW maximum pre-repair per-hour dispatch-minus-load mismatch. These diagnostics never project, repair, or replace a state for hardware credit.

5. Matched baselines and platform-native resources

Frozen objective	Matched classical reference	Dirac-3 representation	Samples / measured seconds	Result
IEEE-33 Stage 1	Exact enumeration, 2,048 states	11 two-level integer variables, degree 2	25 / 6 s	Exact
IEEE-33 Stage 2	Exact enumeration, 8 states	3 two-level integer variables, degree 2	25 / 4 s	Exact
IEEE-33 Stage 3 qudit	Exact enumeration, 3 states	1 ternary qudit, degree 3	25 / 4 s	Exact
IEEE-33 Stage 3 continuous	Deterministic one-dimensional oracle	2 continuous, sum=1, degree 3	25 / 4 s	0.00378% gap
IEEE-39 hourly	Converged SLSQP, identical polynomial	3-4 continuous, sum=10, degree 2	4 x 25 / 19 s	0.00606%-1.728% gaps
IEEE-39 window (4 h)	Analytic convex-simplex KKT oracle	13 continuous, sum=40, degree 2	25 / 9 s	1.039% gap

These are matched-objective comparisons. Exact enumeration solves the 2,048-state Stage-1 instance in 5.6 ms median over ten repeats. The complete credential-free scientific core, excluding dependency installation and figure rendering, completed in 6.2 s on the final local Python 3.12 audit host. The measured platform value is direct higher-order, qudit, and simplex execution without gate compilation or binary quadratization. Dirac-3 is an analog qudit optimizer, so gate-model qubit count, circuit depth, and shots are not applicable. Platform-native accounting uses variable/qudit count, levels or sum constraint, polynomial degree, schedule, returned samples, coefficient resolution, and device seconds. The nine in-spec jobs returned 225 samples in 46 device seconds. The largest in-spec job has 13 continuous variables at degree 2; native-cubic cases use one ternary qudit or two continuous variables. We claim platform-native representational benefit and verified solution quality, not computational speedup.

6. Reproducibility, limitations, and conclusion

A clean Python 3.11-3.12 environment runs `python run_all_local.py` with pinned direct dependencies. It regenerates both instances, 27 feeder payloads, coefficient audits, baselines, the AC screen, chronology cases, 15 IEEE-39 convention/invariant gates, 10 feeder tests, 11 release/evidence tests, and 39 consolidated claims. The release manifest is verified before any output is regenerated. The source distinguishes static critical shortfall (MWh), expected annual critical energy not served (MWh/year), customer-block fraction, and load-weighted system fraction. The public sources are PYPOWER case39 [4] and MATPOWER case33bw [3]. No credential is stored.

The method uses Dirac-3 because the QCi track requires working QCi execution and the dispatch objectives benefit from native higher-order polynomials, ternary qudits, and continuous simplexes. Re-binarizing the cubic terms would add precision bits and quadratization variables. The evidence does not establish utility deployment, transient stability, protection coordination, inverter reactive limits, or asymptotic scaling. It establishes a compact, fully inspectable result whose fixed planning decision, machine feasibility, classical references, physical interpretation, and limitations survive independent rerun.

IP and licensing boundary.

Citations do not supply external source code or omitted background implementation. Submitted material remains governed by the applicable Challenge rules, participant agreement, and Aqora terms; no additional general-public licence is granted. Certain subject matter in and related to the submission is disclosed in pending Australian provisional patent applications; no granted patent is claimed. Licensing enquiries for rights not already granted under the competition instruments are listed in the README.

References

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